WEEK 1: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Foundation of Pre-Technical Studies

Sub Strand: Safety on Raised Platforms

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Define the term "raised platforms".
2. Identify the types of raised platforms used in a workplace.
3. Discuss the uses of different types of raised platforms in a workplace.
4. Appreciate the importance of safety when using raised platforms.

Key Inquiry Questions:

- What is a raised platform in a workplace?
- What are the uses of raised platforms in a workplace?

Learning Resources:

- Learner's book
- Lesson notes
- Digital devices (tablets/computers)
- Pictures of raised platforms
- Video clips demonstrating different types of raised platforms and their uses

Organisation of Learning:

Introduction (5 minutes):

- Briefly review the previous lesson on workplace safety.

- Present the topic for today and its relevance to real-life work environments.

- Engage students in a discussion on safety and its importance when working at heights.

Lesson Development (30 minutes):

Step 1: Define Raised Platforms

- In pairs, learners brainstorm the definition of "raised platform".

- Guide a class discussion to consolidate ideas and agree upon a clear definition.

Step 2: Identify Types of Raised Platforms

- Display pictures of various raised platforms such as ladders, scaffolding, and elevated work platforms.

- Students identify and categorize the types of raised platforms they see.

- Discuss the features that make each type suitable for different tasks.

Step 3: Explore Uses of Raised Platforms

- Using digital devices, students search for information on the uses of the identified raised platforms in workplaces (e.g., construction sites, warehouses).

- Groups share their findings, focusing on how each type of platform enhances safety and efficiency.

Step 4: Discuss Safety Considerations

- Lead a discussion on the importance of safety procedures when using raised platforms (e.g., wearing harnesses, proper training).

- Encourage students to reflect on how different platforms have specific safety requirements.

Conclusion (5 minutes):

- Summarize the key points: definition of raised platforms, types identified, and their uses.

- Conduct a quick interactive quiz to reinforce key concepts learned during the lesson.

- Briefly introduce the next lesson topic, piquing students' curiosity with questions such as "What safety equipment is essential when working on raised platforms?"

Extended Activities:

- Research Project: Create a brochure or presentation detailing a specific type of raised platform, including its uses and safety measures associated with it.

- Field Trip: If possible, arrange a visit to a local construction site to observe raised platforms in use and discuss safety practices encountered there.

- Safety Poster Competition: Design a poster demonstrating safety tips when working on raised platforms, which can be displayed in the classroom or school.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 1: LESSON 2

Strand: Foundation of Pre-Technical Studies

Sub Strand: Safety on Raised Platforms

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. State the risks associated with working on raised platforms.
2. Describe the risks associated with working on raised platforms.
3. Search the internet for clips on the risks associated with working on raised platforms.
4. Acknowledge the risks associated with working on raised platforms.

Key Inquiry Question(s):

- What are the risks associated with working on raised platforms?

Learning Resources:

- Learner's book
- Lesson notes
- Digital devices
- Video clips
- Pictures

Organisation of Learning:

Introduction (5 minutes):

1. Review the previous lesson: Ask students to share what they learned in the last class, prompting discussion about safety measures in various environments.

2. Discussion of Key Concepts: Guide students to read through relevant content in their learner's book and discuss it with peers. Emphasize understanding key risks involved in working at heights.

Lesson Development (30 minutes):

Step 1: Brainstorming Session

- Activity: In groups, students brainstorm and list potential risks they think are involved in working on raised platforms (e.g., falling, equipment failure).

- Outcome: Each group shares their list with the class, highlighting commonalities.

Step 2: Research and Exploration

- Activity: Using their digital devices, students search for internet clips and images illustrating the risks identified. They can use guided prompts like "safety risks working on scaffolding" or simply "hazards on raised platforms."

- Outcome: Groups will compile findings and prepare to present them to the class.

Step 3: Group Discussion

- Activity: Facilitate a class discussion where groups talk about what they discovered in their research. Encourage them to explain how those risks could impact workers or safety measures that could mitigate those risks.

- Outcome: Promote critical thinking and communication skills through shared insights.

Step 4: Poster Preparation

- Activity: Each group prepares a poster showcasing the risks of working on raised platforms, emphasizing visuals and clear descriptions of each risk.

- Outcome: Posters will serve as a valuable visual tool that can be used in future discussions.

Conclusion (5 minutes):

- Key Points Summary: Recap the important risks discussed and the role of safety measures when working at heights.

- Interactive Activity: Conduct a quick quiz or a game that involves matching risks to their proper safety measures to reinforce understanding.

- Preview Next Session: Briefly introduce the upcoming lesson on safety equipment used on raised platforms. Pose questions for them to think about: "What safety equipment do you think is essential when working at heights?"

Extended Activities:

- Research Project: Assign students to independently research a specific safety regulation related to raised platforms and present their findings in the next class.

- Classroom Safety Audit: Organize a classroom activity where students identify potential risks in their learning environment related to the principles they've learned and suggest improvements.

- Guest Speaker: Invite a safety expert or construction worker to speak about the real-world application of safety measures and protocols when working at heights.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 1: LESSON 3

Strand: Foundation of Pre-Technical Studies

Sub Strand: Safety on Raised Platforms

Specific Learning Outcomes:

- By the end of the lesson, students should be able to:

1. Identify safety measures to observe when working on raised platforms in the workplace.
2. Discuss these safety measures with their peers.
3. Prepare flashcards or posters detailing the safety measures.
4. Acknowledge the importance of these safety precautions.

Key Inquiry Question:

- What safety measures should one observe when working on raised platforms in a workplace?

Learning Resources:

- Learner's book
- Lesson notes
- Digital devices (tablets/laptops)
- Posters
- Marker pens
- Video clips on safety protocols

Organisation of Learning:

Introduction (5 minutes):

- Begin with a quick review of the previous lesson, focusing on workplace safety in general.

- Ask students to share any safety practices they are familiar with. This could lead into the topic at hand.

- Guide learners to read and discuss key concepts related to safety measures for raised platforms from their books.

Lesson Development (30 minutes):

Step 1: Brainstorming Session

- In small groups, students brainstorm on ways to minimize risks when working on raised platforms.

- Encourage them to think of practical solutions and share experiences if applicable.

Step 2: Viewing and Discussion

- Show a video clip that illustrates safety measures for working on raised platforms.

- After viewing, facilitate a group discussion where students can reflect on the video and share insights on what they observed.

Step 3: Collaborative Poster Preparation

- Assign each group to create a poster that highlights the key safety measures they discussed.

- Provide materials like markers and large sheets of paper for them to use. Each group will present their poster to the class afterward.

Step 4: Presentation and Reflection (Optional, if time allows)

- Have each group briefly present their poster and explain one or two safety measures in detail.

- Encourage questions and comments from peers to foster collaboration and deeper understanding.

Conclusion (5 minutes):

- Summarize key points discussed in the lesson, highlighting the importance of safety on raised platforms.

- Conduct a brief interactive activity, such as a quiz or a few quick questions, to reinforce the safety measures learned.

- Preview the next session's topics, encouraging students to consider how safety practices are implemented in their daily lives or specific jobs they might be interested in.

Extended Activities:

- Safety Simulation: Create a simulated work environment with raised platforms where students can role-play and practice safety measures.

- Research Assignment: Have students research different industries that use raised platforms and report on the specific safety regulations they must follow.

- Flashcard Creation: As homework, students can create flashcards for each safety measure and quiz their family members on the importance of these precautions.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 1: LESSON 4

Strand: Foundation of Pre-Technical Studies

Sub Strand: Safety on Raised Platforms

Specific Learning Outcomes:

- By the end of the lesson, the learner should be able to:

1. State the need for observing safety while working on raised platforms in workplaces.

2. Discuss the importance of observing safety when working on raised platforms.

3. Prepare flashcards showing the importance of observing safety when working on raised platforms.

4. Appreciate the need for observing safety while working on raised platforms.

Key Inquiry Question:

- What is the importance of observing safety when working on raised platforms?

Learning Resources:

- Learner's book

- Digital devices (for research)

- Flashcards

- Lesson notes

Organisation of Learning:

Introduction (5 minutes):

- Briefly review the previous lesson on workplace safety.

- Guide learners to read relevant content from the learner's book, focusing on safety measures for working on raised platforms.

- Discuss key concepts as a class to ensure understanding.

Lesson Development (30 minutes):

Step 1: Brainstorming Session

- In small groups, students will brainstorm the potential hazards of working on raised platforms.

- Each group will then present their findings to the class.

Step 2: Discussion of Importance

- Engage the class in a whole-group discussion about the importance of safety measures when working on raised platforms.

- Prompt students to think about real-life examples where safety is critical, such as construction sites.

Step 3: Creating Flashcards

- Still in groups, learners will create flashcards that highlight key safety practices and their importance.

- Encourage creativity and clarity in their designs.

Step 4: Presenting Flashcards

- Each group will briefly present their flashcards to the class, explaining the safety practices they have highlighted.

Conclusion (5 minutes):

- Summarize the key points discussed about safety on raised platforms.

- Conduct a quick interactive quiz where learners can answer questions based on the lesson to reinforce understanding.

- Prepare learners for the next session by previewing topics such as different types of raised platforms and specific safety equipment used.

Extended Activities:

- Safety Poster Project: Students can design a safety poster that highlights the key points discussed in class to be hung in the school or shared within the community.

- Field Trip: Plan a visit to a construction site or local business that utilizes raised platforms, allowing students to observe safety practices in a real-world setting.

- Safety Protocol Role-Play: In groups, students can enact scenarios demonstrating both unsafe and safe practices while working on raised platforms, fostering discussion about the outcomes.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 2: LESSON 1

Strand: Foundation of Pre-Technical Studies

Sub Strand: Safety on Raised Platforms

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Role play safety practices while working on raised platforms.

2. Enjoy role playing the safety practices while working on raised platforms.

Key Inquiry Question:

- How do we ensure safety when working on raised platforms?

Learning Resources:

- Digital devices

- Open area

- Raised platforms (e.g., ladders)

Organisation of Learning:

Introduction (5 minutes):

- Review: Begin with a quick recap of the previous lesson focusing on general safety guidelines in technical studies.

- Discussion: Ask students to share any experiences relating to safety practices. Link their experiences to the importance of safety on raised platforms.

Lesson Development (30 minutes):

Step 1: Group Formation

- Divide the class into small groups of 4-5 students.

- Assign each group to a specific safety topic related to working on raised platforms (e.g., proper ladder setup, wearing safety gear, maintaining three points of contact, and emergency procedures).

Step 2: Research and Planning

- Each group uses digital devices to research their assigned topic.

- Encourage students to note key points and decide how they will present their safety practice through role play. Remind them to focus on clear and actionable safety practices.

Step 3: Role Playing

- Groups perform their role plays in the open area.

- As they present, remind them to demonstrate effective communication and safety practices illustrated in their research.

Step 4: Feedback and Reflection

- After each group's presentation, allow peers to give constructive feedback on the role plays.

- Facilitate a brief discussion on how the role plays can be applied in real-life situations when working on raised platforms.

Conclusion (5 minutes):

- Summary: Recap the key safety practices discussed and how they help prevent accidents when working at heights.

- Interactive Activity: Conduct a quick quiz or game where students identify correct and incorrect safety practices based on scenarios you present.

- Preview: Give learners a sneak peek of the next lesson, which will cover risk assessments in technical studies, and pose the question: "What could go wrong if we don't follow safety protocols?"

Extended Activities:

- Research Assignment: Ask students to create a safety poster or video demonstrating the importance of safety when using raised platforms. Display these around the classroom or school.

- Home Connection: Encourage learners to discuss safety practices with family members who work in construction, maintenance, or other fields involving raised platforms, and report back on their findings.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 2: LESSON 2

Strand: Foundation of Pre-Technical Studies

Sub Strand: Safety on Raised Platforms: Assessment

Specific Learning Outcomes:

By the end of the lesson, the learner should be able to:

1. Attempt questions on the sub-strand of safety on raised platforms.

Key Inquiry Question(s):

- What are the potential hazards associated with working on raised platforms?

- How can we ensure safety when using raised platforms in various environments?

Learning Resources:

- Learner's Book

- Teacher's Assessment Questions

Organisation of Learning:

Introduction (5 minutes):

- Begin with a brief review of the previous lesson on the importance of safety protocols in different environments, including raised platforms.

- Guide learners to read a relevant section from the Learner's Book that discusses safety measures when working on raised platforms. Encourage a short discussion to clarify key concepts such as guardrails, harnesses, inspection of equipment, and awareness of surroundings.

Lesson Development (30 minutes):

Step 1: Define Safety Measures

- Discuss the fundamental safety measures required when working on raised platforms such as wearing protective gear, using harnesses, and conducting site inspections.

- Have students list these measures on the whiteboard and discuss real-life examples where these measures might be necessary.

Step 2: Identify Risks

- In pairs, learners will brainstorm potential risks associated with raised platforms. They can list these risks in their exercise books.

- After 5 minutes, facilitate a sharing session where groups present their findings. Discuss common themes and unique risks they identified.

Step 3: Assessment Questions

- Distribute Teacher's Assessment Questions based on the sub-strand. These questions should challenge them to use the knowledge gathered in the previous steps and apply it to hypothetical scenarios.

- Allow students 5 minutes of quiet time to attempt the questions individually.

Step 4: Discuss Answers

- Review the answers to the assessment questions as a class. Encourage students to explain their reasoning. This will help reinforce their understanding and allow them to learn from each other's perspectives.

Conclusion (5 minutes):

- Summarize the key points discussed during the lesson, especially the importance of safety protocols and identifying risks involved in working on raised platforms.

- Conduct a brief interactive activity where students can share one safety measure and one risk they learned today.

- Preview the next lesson, which will explore the design and construction of safer raised platforms.

Extended Activities:

- Research Project: Assign students to research different types of raised platforms used in construction and other fields, focusing on specific safety protocols in place for each type.

- Safety Campaign: Students can work in groups to create a short presentation or poster campaign promoting safety regulations for raised platforms. They can present this to the class or prepare it for display in a common area of the school.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 2: LESSON 3

Strand: Foundation of Pre-Technical Studies

Sub Strand: Self-Exploration and Career Development

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the ways of nurturing talents and abilities for business purposes.
2. Explain the ways of nurturing talents and abilities for business purposes.
3. Conduct online research about nurturing talents and abilities for business purposes.
4. Acknowledge the various methods used in society to nurture talents and abilities.

Key Inquiry Questions:

- What does it mean to nurture a talent and ability?
- How are talents and abilities nurtured?

Learning Resources:

- Lesson notes
- Digital devices
- Learner's Book
- Posters and Flashcards

Organisation of Learning:

Introduction (5 minutes):

- Begin with a quick recap of the previous lesson focused on the importance of skills and abilities in our everyday lives.

- Guide the learners to read and discuss relevant content from available resources, ensuring that they understand the key concepts of nurturing talents and abilities.

Lesson Development (30 minutes):

Step 1: Understanding 'Nurture'

- In small groups, learners will use dictionaries to find and discuss the definition of "nurture."

- Encourage each group to share their interpretations and summarize their findings.

Step 2: Researching Nurturing Techniques

- Learners will utilize digital devices to research various ways to nurture talents and abilities for business purposes.

- Provide guiding questions to help their research focus, such as:

- What activities or resources help develop business skills?

- Who in the community supports talent development?

Step 3: Discussion on Nurturing in the Community

- Groups will discuss within their teams the different ways talents and abilities are nurtured in their own communities.

- Each group should prepare to share one unique method they discovered.

Step 4: Creating Visual Aids

- Using materials provided, learners will create posters or flashcards that illustrate the ways of nurturing talents and abilities.

- They can draw pictures, write concepts, or even add quotes from their research to make it engaging.

Conclusion (5 minutes):

- Summarize the key points discussed during the lesson focusing on definitions, methods of nurturing, and community resources.

- Engage students in a quick interactive activity, such as a "think-pair-share," where they share one method of nurturing talents with a partner.

- Preview the next session by introducing topics related to entrepreneurship or personal finance, encouraging students to think about how they can apply their talents in a business setting.

Extended Activities:

- Talent Showcase: Organize a session where learners can present a talent they possess and discuss how they nurture it.

- Research Assignment: Assign learners to select a local business they admire and investigate how the owner nurtured their skills to succeed in that business.

- Mentorship Project: Pair learners with local professionals for short mentorship experiences to see how skills are developed in real-world situations.

Teacher Self-Evaluation:

WEEK 2: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Foundation of Pre-Technical Studies

Sub Strand: Self-Exploration and Career Development

Specific Learning Outcomes

- By the end of the lesson, learners should be able to:

1. Define the terms "self-exploration" and "career development."

2. Discuss the importance of self-exploration for career development.

3. Search the internet for information on the importance of self-exploration for career development.

4. Appreciate the need for self-exploration for career development.

Key Inquiry Question:

- Why is self-exploration necessary for career development?

Learning Resources:

- Lesson notes

- Digital devices (tablets, laptops, etc.)

- Learner's book

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson: Start with a quick recap of what was learned in the last class. Engage students by asking questions related to the previous topic.

- Reading and Discussion: Guide learners to read a passage from their lesson notes regarding self-exploration and career development. Facilitate a discussion by asking them to explain these concepts in their own words.

Lesson Development (30 minutes):

Step 1: Define Key Terms

- Activity: In groups, learners will look up the definitions of "self-exploration" and "career development" using digital devices or print resources.

- Group Work: Each group will write down their definitions on a shared board for everyone to see and discuss.

Step 2: Discuss Your Findings

- Activity: Groups will share their definitions and discuss why self-exploration is important for career development.

- Class Participation: Encourage students to voice their opinions and ask questions about each group's findings. Highlight key points.

Step 3: Research the Importance

- Research Task: Learners will take a few minutes to search online for articles or videos that explain the importance of self-exploration in career development.

- Sharing Findings: Groups will report back on what they found and how it relates to personal experiences or future goals.

Step 4: Talents and Abilities Showcase

- Discussion: Encourage students to think about how they can showcase their talents and interests through clubs, societies, or other activities at school.

- Brainstorming: As a class, brainstorm some extracurricular activities that align with students' interests and career aspirations.

Conclusion (5 minutes):

- Summarize: Recap the definitions of self-exploration and career development, and discuss why self-exploration is essential.

- Interactive Activity: Conduct a brief quiz or game where students match definitions to the correct terms or complete sentences related to the importance of self-exploration.

- Preview of Next Lesson: Introduce the upcoming topics or questions they should think about at home related to their own career interests.

Extended Activities:

- Self-Exploration Journal: Encourage students to keep a journal where they write about their interests, experiences, and career aspirations. They can reflect on what they discover about themselves each week.

- Guest Speaker: Arrange for a local professional to come and share their career journey, emphasizing the role of self-exploration in their success.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 3: LESSON 1

Strand: Foundation of Pre-Technical Studies

Sub Strand: Self-Exploration and Career Development

Specific Learning Outcomes:

- By the end of the lesson, the learner should be able to:

1. Identify the ethical and unethical practices related to the use of talents and abilities.
2. Analyze these practices for business purposes.
3. Prepare educational posters on these practices.
4. Acknowledge the impact of ethical and unethical practices on our lives.

Key Inquiry Question:

- What are the ethical and unethical practices related to the use of talents and abilities?

Learning Resources:

- Case Studies
- Lesson Notes
- Digital Devices
- Learner's Book

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson briefly, focusing on what talents and abilities mean.

- Guide learners to explore and discuss the relevant content from the learning resources, highlighting ethical and unethical practices.

Lesson Development (30 minutes):

Step 1: Definition and Discussion

- In groups, students will clarify the meaning of "ethical" and "unethical" in the context of talents and abilities.

- Groups share their understanding with the class, fostering discussion on differing viewpoints.

Step 2: Case Study Examination

- Distribute a case study that illustrates both ethical and unethical practices in a business setting.

- Groups read and discuss the case, identifying specific examples of both practices.

Step 3: Classification Activity

- Each group lists at least three ethical practices and three unethical practices based on their understanding and the case study.

- They will then prepare to share these classifications with the class.

Step 4: Poster Preparation

- Groups create educational posters that convey messages about the ethical and unethical practices related to talents and abilities.

- Encourage creativity in their presentations—these can include drawings, quotations, and short explanations.

Conclusion (5 minutes):

- Summarize the key points learned regarding ethical and unethical practices.

- Engage students in an interactive activity, such as a quick quiz on concepts discussed.

- Preview the next session's topics and encourage students to think about how these practices affect their daily lives.

Extended Activities:

- Reflection Journal: Students write a short reflection on a time they witnessed ethical or unethical practices in their lives.

- Research Project: Choose a famous person and research examples of how they used their talents ethically or unethically in their career.

- Debate: Organize a class debate on the importance of ethics in the business world, focusing on how talents and abilities can impact ethical decision-making.

Teacher Self- Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 3: LESSON 2

Strand: Foundations of Pre-Technical Studies

Sub Strand: Self-Exploration and Career Development

Specific Learning Outcomes:

- By the end of the lesson, learners will be able to:

1. Differentiate between talents and abilities.
2. Identify the different talents and abilities that individuals may possess.
3. Search the internet for examples of talents and abilities an individual may have.
4. Acknowledge the different talents and abilities an individual may possess.

Key Inquiry Questions:

- What is the difference between talents and abilities?
- What are some examples of talents and abilities that one can possess in the community?

Learning Resources:

- Learner's Book
- Lesson notes
- Flashcards
- Digital devices

Organisation of Learning:

Introduction (5 minutes):

- Briefly review the previous lesson to connect to today’s topic.

- Introduce the key concepts of talents and abilities, encouraging students to share any initial thoughts.

Lesson Development (30 minutes):

Step 1: Brainstorming Session

- In small groups, have learners discuss what they believe are the differences between talents and abilities.

- Each group will create a list of definitions and examples to share with the class.

Step 2: Group Presentations

- Groups will take turns presenting their definitions and examples of talents and abilities.

- Facilitate a class discussion, asking for clarification or expansion on the ideas presented.

Step 3: Research Activity

- Students will use digital devices individually or in pairs to research the difference between talents and abilities.

- Encourage learners to find reliable sources and jot down key points that help clarify the definitions.

Step 4: Flashcard Creation

- Using the findings from their research, students will prepare flashcards that illustrate different talents and abilities.

- Each card should include a definition, example, and image related to the talent or ability.

Conclusion (5 minutes):

- Summarize the key points from the lesson, highlighting the distinctions between talents and abilities.

- Conduct a brief interactive quiz where students match talents to abilities, reinforcing understanding.

- Preview the next session, hinting at how talents and abilities relate to future career paths.

Extended Activities:

- Talent Show: Organize a classroom talent show where students can showcase their unique talents or abilities, fostering a sense of community and self-confidence.

- Research Projects: Assign students to pick a career that interests them and research the talents and abilities needed for that field, culminating in a presentation to the class.

- Reflection Journals: Have students maintain a reflection journal throughout the week, noting any talents or abilities they see in themselves or in their peers.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 3: LESSON 3

Strand: Foundations of Pre-Technical Studies

Sub Strand: Self-Exploration and Career Development

Specific Learning Outcomes:

- By the end of the lesson, learners will be able to:

1. Identify the pathways in senior school.

2. Relate different talents and abilities to career pathways in senior school.

3. Prepare a PowerPoint presentation showing a list of talents and abilities corresponding to career pathways in senior school.

4. Acknowledge how different talents and abilities correspond to career pathways in senior school.

Key Inquiry Question(s):

- What are the career pathways in the senior school?

Learning Resources:

- Learner's books

- Chart showing career pathways in senior school

- Lesson notes

- Digital devices (computers/tablets)

- Flashcards with different talents and abilities

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson about the importance of planning for the future and how interests can inform career choices.

- Guide learners to revisit key concepts by reading excerpts from the learning resources, highlighting the significance of understanding career pathways in shaping their educational journey.

Lesson Development (30 minutes):

Step 1: Explore Career Pathways

- In groups, learners observe the chart displaying different career pathways available in senior school.

- Each group discusses the pathways they see and what they might entail.

Step 2: Identify Talents and Abilities

- Using flashcards with various talents and abilities, groups will match these to the career pathways identified in Step 1.

- Encourage students to consider how their own interests and skills fit into these pathways.

Step 3: List Creation & PowerPoint Preparation

- Each group collaborates to create a comprehensive list of talents/abilities and their corresponding career pathways.

- Students begin preparing a PowerPoint presentation that will showcase their findings.

Step 4: Presentation Preparation

- Groups finalize their PowerPoint presentations, ensuring that their lists are clear and visually appealing.

- Briefly discuss how they will present their findings to the class in the next lesson.

Conclusion (5 minutes):

- Summarize the key points discussed, confirming what career pathways are and how talents and abilities shape career choices.

- Conduct a quick interactive quiz or KWL (Know, Wonder, Learned) activity to reinforce the material.

- Preview the next session, where groups will present their PowerPoint to the class and receive feedback.

Extended Activities:

- Talent Inventory Assignment: Learners can create an inventory of their own talents and abilities for a homework assignment, researching potential career pathways that match their interests.

- Job Shadowing Project: Encourage students to find local professionals in fields of interest and reach out to them for job shadowing or informational interviews.

- Guest Speaker Session: Organize a session with speakers from various careers to talk about their pathways and how they aligned with their abilities.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 3: LESSON 4

Strand: Foundations of Pre-Technical Studies

Sub-Strand: Self-Exploration and Career Development

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the career opportunities related to talents and abilities in Pre-Technical Studies.

2. Interview a resource person about career opportunities in Pre-Technical Studies.

3. Acknowledge the importance of these career opportunities in relation to their own talents and abilities.

Key Inquiry Question:

- What are the career opportunities related to talents and abilities in Pre-Technical Studies?

Learning Resources:

- Resource person

- Digital devices

- Learner's book

- Lesson notes

Organisation of Learning:

Introduction (5 minutes):

- Begin class with a quick review of the previous lesson, asking students to share what they learned.

- Guide learners to read a relevant excerpt from their textbooks that discusses careers in Pre-Technical Studies. Encourage a brief discussion to establish understanding of the key concepts.

Lesson Development (30 minutes):

Step 1: Introduction to Careers

- Introduce the resource person (a professional in a relevant field).

- Prepare students with a few questions they might want to ask, such as:

- What inspired you to choose this career?

- What skills are most important for success in your field?

Step 2: Group Interaction with Resource Person

- Divide the class into small groups.

- Each group will have the opportunity to interview the resource person. Encourage each group to take notes on answers to the questions prepared in Step 1, as well as any interesting follow-up insights shared.

Step 3: Group Discussion

- After the interviews, reconvene as a class.

- Guide a discussion where students share what they learned about the different career opportunities related to their talents and abilities.

- Facilitate conversation on how these careers link to the skills developed in Pre-Technical Studies.

Step 4: Wrap-up Activity

- Ask learners to write down one career they found particularly interesting and one skill they think they need to develop further based on the discussion.

Conclusion (5 minutes):

- Summarize the key points from the lesson: highlights of career opportunities related to Pre-Technical Studies and the importance of aligning personal talents with career choices.

- Conduct a quick interactive activity: ask students to share their written insights with a partner, fostering engagement.

- Provide a preview of the next lesson, possibly introducing a related topic like the skills required for success in various technical careers.

Extended Activities:

- Research Project: Have students choose a specific career in Pre-Technical Studies and create a presentation about the required skills, educational pathways, and potential job market.

- Guest Speaker Reflection: After the resource person visit, students write a reflection on what they learned and how it impacts their view of their own career aspirations.

- Career Fair Preparation: Organize a class project to host a mini-career fair where students present on various careers related to Pre-Technical Studies.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 4: LESSON 1

Strand: Foundations of Pre-Technical Studies

Sub Strand: Handling Hazardous Substances

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify hazardous substances found in the environment.
2. Use digital devices to search for information on hazardous substances.
3. Draw the symbols of hazardous substances.
4. Acknowledge the hazardous substances in the environment.

Key Inquiry Questions:

- What are hazardous substances in the environment?

Learning Resources:

- Lesson notes
- Digital devices (tablets/laptops)
- Learner's Books
- Charts (hazard symbols)
- Pictures illustrating hazardous substances

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson on safety in the environment.

- Guide learners to read and discuss relevant content from their books, focusing on the understanding of hazardous substances.

Lesson Development (30 minutes):

Step 1: Brainstorming

- In groups of 4-5, learners will brainstorm what they think hazardous substances are. Encourage them to think of examples they have seen or heard about in their daily lives (e.g., chemicals at home, industrial waste).

- Groups share their ideas, and the teacher can compile a list on the board to foster discussion.

Step 2: Study Hazard Symbols

- Introduce the symbols for poisonous, flammable, and corrosive substances using charts and digital resources.

- Discuss the meaning and examples of each symbol, encouraging learners to ask questions if they don’t understand something.

Step 3: Research Activity

- Using digital devices, guide learners to conduct research on hazardous substances in their local environment. They can look for specific substances used in industry, household products, or common pollutants.

- Each group should gather at least three facts about a specific substance and prepare to present their findings.

Step 4: Drawing Activity

- Have learners draw the symbols of the poisonous, flammable, and corrosive substances they studied earlier. They can use the charts as a reference.

- Ask them to label their drawings with a brief description of what each symbol means.

Conclusion (5 minutes):

- Summarize the lesson by highlighting key points and the outcomes achieved, particularly the identification and recognition of hazardous substances and symbols.

- Conduct a brief interactive quiz using questions about the hazardous substances discussed to reinforce learning.

- Prepare learners for the next session by previewing upcoming topics, such as the impact of hazardous substances on health and the environment.

Extended Activities:

- Create a Safety Poster: Students can create a poster with drawings and information about one specific hazardous substance, including its symbol, effects, and safety measures. Display these in the classroom or school.

- Community Survey: Learners can take a survey of harmful substances at home (with parental consent) or in their community and report their findings in the next class.

- Guest Speaker: Arrange for a local environmental scientist or safety officer to talk to the class about hazardous substances and their management.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 4: LESSON 2

Strand: Foundations of Pre-Technical Studies

Sub Strand: Handling Hazardous Substances

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Classify the hazardous substances found in the locality.

2. Use digital devices to prepare a table showing the classification of the hazardous substances.

3. Enjoy classifying different hazardous substances found in the locality.

Key Inquiry Question:

- Which hazardous substances are found in the locality?

Learning Resources:

- Lesson notes on hazardous substances in the environment

- Digital devices (tablets/laptops)

- Learner's book

Organisation of Learning:

Introduction (5 minutes):

- Review Previous Lesson: Begin with a quick recap of what was learned in the last lesson and its relevance to today's topic.

- Reading and Discussion: Guide learners to read selected passages from their lesson notes regarding hazardous substances. Initiate a discussion to ensure they understand the definitions and importance of recognizing these substances.

Lesson Development (30 minutes):

Step 1: Group Discussion

- Divide learners into small groups. Ask each group to brainstorm and list hazardous substances they have observed in their locality. Provide examples, such as household cleaners, pesticides, or chemicals used in local industries.

Step 2: Classification Activity

- Groups will then categorize their substances into three classifications:

- Poisonous (e.g., rat poison)

- Corrosive (e.g., drain cleaners)

- Flammable (e.g., gasoline)

- Each group should discuss the properties of each category and why items belong to a specific classification.

Step 3: Digital Table Preparation

- Using digital devices, each group will create a table to present their findings. The table should have columns for the substance name, category (poisonous, corrosive, flammable), and a brief description of each.

Step 4: Group Presentations

- Each group presents their table to the class. Encourage other students to ask questions or add any other hazardous substances they might know about.

Conclusion (5 minutes):

- Summary: Recap the types of hazardous substances discussed, emphasizing the significance of understanding them.

- Interactive Activity: Conduct a short game where students match hazardous substances to their classifications using flashcards, reinforcing their learning.

- Preview Next Session: Briefly introduce the upcoming topic related to safety precautions when handling hazardous substances and possible experiments related to this.

Extended Activities:

- Research Assignment: Have students choose one hazardous substance from their classification and prepare a short report that includes its uses, risks, and safety measures.

- Local Field Study: Organize a field trip (or a virtual tour) to a local waste disposal site or industry to learn how hazardous materials are managed in their area.

- Create Awareness Materials: Encourage students to design posters or digital presentations that raise awareness about the safe handling of hazardous substances.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

WEEK 4: LESSON 3

Strand: Foundations of Pre-Technical Studies

Sub Strand: Handling Hazardous Substances

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1.Identify safe ways of handling hazardous substances in the environment.

2. Describe safe ways of handling hazardous substances in the environment.

3. Use digital devices to search for information and clips on safe ways of handling hazardous substances in the environment.

4. Acknowledge the importance of safe handling practices.

Key Inquiry Question(s):

- How are hazardous substances handled?

- Why are hazardous substances labelled?

Learning Resources:

- Lesson notes

- Learner's book

- Samples of hazardous substances (e.g., cleaning supplies, chemicals)

- Digital devices (tablets/computers)

- Posters illustrating safety procedures

Organisation of Learning:

Introduction (5 minutes):

- Begin the lesson by briefly reviewing the previous lesson's content.

- Prompt students with questions to activate prior knowledge, such as “What do you know about hazardous substances?”

- Guide learners to read and discuss key concepts from the learning resources, focusing on understanding the terminology and definitions relevant to hazardous substances.

Lesson Development (30 minutes):

Step 1: Research

- Divide the class into small groups.

- Assign each group the task of using digital devices to research “Safe Handling of Hazardous Substances.” Encourage them to find reliable websites, articles, and video clips that detail proper handling techniques.

Step 2: Group Discussion

- Each group discusses their findings, focusing on safe practices and why they are important.

- Encourage groups to consider questions like, "What could go wrong if hazards are not handled properly?"

Step 3: Reading Instructions

- Distribute printed instructions or safety data sheets about common hazardous substances.

- Each group reads and interprets the instructions, discussing key points such as storage, personal protective equipment (PPE), and emergency measures.

Step 4: Create Posters

- Groups design informative posters that showcase the safe ways of handling hazardous substances, highlighting the key points discussed.

- Include visuals, such as icons or illustrations, to make the information clear and engaging.

Conclusion (5 minutes):

- Summarize the key points and learning objectives achieved during the lesson.

- Conduct an interactive activity featuring a quick quiz or “true or false” game to reinforce what students learned about handling hazardous substances safely.

- Preview the next session by introducing the upcoming topic: “Understanding Labels on Hazardous Materials.”

Extended Activities:

- Research Project: Assign a project where students choose a hazardous substance and create a presentation or report detailing its proper handling, potential risks, and emergency procedures.

- Safety Audit: Invite students to conduct a safety audit of their homes or classrooms, identifying any hazardous substances and suggesting safe handling practices they could implement.

- Guest Speaker: Arrange for a visit from a local health and safety officer or environmental scientist to speak about real-life applications of hazardous substance handling.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL

	GRADE 9	**PRE - TECHNICAL STUDIES**			

WEEK 4: LESSON 4

Strand: Foundations of Pre-Technical Studies

Sub Strand: Handling Hazardous Substances

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Handle hazardous substances safely in the environment.

2. Adhere to safety protocols when handling any hazardous substance.

Key Inquiry Question:

- How are hazardous substances handled?

Learning Resources:

- Lesson notes

- Samples of hazardous substances (if safe and appropriate)

- Digital devices (for recording practice sessions)

Organisation of Learning:

Introduction (5 minutes):

1. Begin with a brief review of the previous lesson, focusing on the importance of safety in various environments.

2. Engage learners in a discussion about hazardous substances, asking them what they know or have heard about them.

3. Guide learners to read and discuss content from the lesson notes, highlighting key safety measures when dealing with hazardous substances.

Lesson Development (30 minutes):

Step 1: Understanding Hazardous Substances

- Introduce different types of hazardous substances:

- Poisonous (e.g., certain plants or chemicals)

- Flammable (e.g., fuels, alcohols)

- Corrosive (e.g., acids)

- Discuss the properties of each type and associated risks.

Step 2: Safety Protocols

- Present safety measures for handling hazardous substances, such as wearing gloves, using goggles, and knowing emergency procedures.

- Conduct a demonstration (if possible) of safe handling practices, highlighting the steps to take in case of accidents.

Step 3: Practical Application

- Split the class into small groups and provide each group with a hypothetical scenario involving hazardous substances (e.g., cleaning a spill, using items in an experiment).

- Ask groups to identify safety measures they would implement in their scenario.

Step 4: Recording and Feedback

- Allow students to use digital devices to record their group discussions or role-plays of safe handling methods.

- Each group presents their recordings to the class, and peers provide constructive feedback on their safety strategies.

Conclusion (5 minutes):

- Recap the key points covered during the lesson, including types of hazardous substances and safety protocols.

- Engage the class in a quick interactive activity, such as a quiz or a round of 'safety scenarios' where they decide the best course of action in response to potential hazards.

- Preview the next session's topic, encouraging students to think about other substances they might encounter in their daily lives.

Extended Activities:

1. Research Project:

Have students research a specific hazardous substance and prepare a presentation or poster that includes its uses, dangers, and safety measures.

2. Field Trip:

Arrange for a visit to a local facility (like a recycling plant or a laboratory) where they handle hazardous materials, and allow them to observe safety measures in practice.

3. Safety Campaign:

Encourage students to create a safety campaign (e.g., posters, videos) promoting safe handling of hazardous substances within their school or community.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL

	GRADE 9	**PRE - TECHNICAL STUDIES**			

WEEK 5: LESSON 1

Strand: Foundations of Pre-Technical Studies

Sub Strand: Handling Hazardous Substances

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. State the importance of observing safety when handling hazardous substances.

2. Explain the importance of observing safety when handling hazardous substances.

3. Appreciate the importance of observing safety when handling hazardous substances.

Key Inquiry Question:

What is the importance of observing safety when handling hazardous substances?

Learning Resources:

- Lesson notes

- Learner's book

- Digital devices

Organisation of Learning:

Introduction (5 minutes):

- Begin the class by quickly reviewing the previous lesson to activate prior knowledge.

- Introduce the day's topic by asking students what they already know about hazardous substances.

- Guide learners to read and discuss relevant content from the learning resources, focusing on key concepts related to safety when handling hazardous substances.

Lesson Development (30 minutes):

Step 1: Brainstorming

- Divide the class into small groups and assign each group the task of brainstorming the potential dangers of handling hazardous substances.

- Ask them to write down as many examples as they can (e.g., chemicals in the lab, household products).

Step 2: Research

- Have each group use digital devices or print resources to research the importance of safety measures when dealing with hazardous substances.

- Encourage them to look for real-world examples or stories illustrating the consequences of neglecting safety.

Step 3: Group Discussion

- Once research is complete, regroup and encourage each group to share their findings with the class.

- Facilitate a class discussion to compare findings and highlight the most critical safety practices.

Step 4: Presentation

- Ask each group to prepare a brief presentation based on their project, including the dangers identified and the safety measures they recommend.

- Allow each group a few minutes to present, focusing on clarity and engagement.

Conclusion (5 minutes):

- Summarize the key points discussed in the lesson, reinforcing the importance of safety in handling hazardous substances.

- Conduct a brief interactive quiz or game to reinforce the main topics, such as identifying safe versus unsafe handling practices.

- Prepare the learners for the next session by previewing upcoming topics or questions to consider.

Extended Activities:

- Safety Poster Project: Students create a safety poster that illustrates important safety measures for handling hazardous substances. This can be displayed around the classroom or school.

- Role-Playing Activity: Students can act out scenarios involving hazardous substances, demonstrating correct and incorrect safety practices.

- Research Assignment: Assign students to investigate a specific hazardous substance, documenting its uses, dangers, and necessary safety precautions. They can present their findings in the next class.

Teacher Self-Evaluation:

WEEK 5: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, the learner should be able to:

1. State the meaning of oblique projection.

2. Identify the characteristics of oblique drawings in technical fields.

3. Use digital devices to search for information on characteristics of oblique drawings.

4. Acknowledge the features of oblique drawings.

Key Inquiry Question(s):

- What are the features of oblique drawings?

Learning Resources:

- Lesson notes

- Digital devices (tablets, laptops)

- Learner's books

- Drawing examples

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson on basic projection techniques.

- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of the key concepts related to oblique projection.

Lesson Development (30 minutes):

Step 1: Introduction to Oblique Projection

- Define oblique projection as a method of drawing objects in a way that shows depth while maintaining the true size of one face.

- Write the definition on the board and explain it using a simple drawing to illustrate the concept.

Step 2: Characteristics of Oblique Drawing

- Group learners into small teams and provide them with digital devices or textbooks.

- Instruct them to research the key characteristics of oblique drawings, such as:

- The angle of the projection (usually 45 degrees)

- The representation of depth

- Distortion in appearance compared to orthographic views.

- Each group will identify at least three characteristics and prepare to share their findings.

Step 3: Group Discussion and Presentation

- Each group presents their findings on the characteristics of oblique drawings.

- Encourage interactive dialogue by inviting other groups to ask questions or add additional insights.

Step 4: Acknowledging Features Through Application

- Provide learners with samples of oblique drawings in your lesson notes.

- Ask students to identify features in these drawings and discuss how they relate to what they learned about the characteristics of oblique projections in their groups.

Conclusion (5 minutes):

- Summarize the key points discussed during the lesson, emphasizing the meaning and characteristics of oblique drawings.

- Conduct a quick interactive quiz using fun questioning (e.g., true or false statements) to reinforce understanding—students can respond using thumbs up or thumbs down.

- Prepare learners for the next session by previewing a topic such as "Applications of Oblique Projection in Engineering," and encourage them to think of where oblique drawings might be used in the real world.

Extended Activities:

-Create Your Own Oblique Drawing: Ask students to choose an object at home (like a book, a toy, or a piece of furniture) and create their own oblique drawing of it. They can use graph paper to help with proportion.

- Classroom Gallery: Organize a classroom exhibition where students can display their oblique drawings and explain the features they used.

- Research Project: Assign students to research different fields (like architecture or game design) where oblique projections are used, and present their findings to the class.

Teacher Self-Evaluation:

WEEK 5: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, students should be able to:

1. Differentiate between cavalier and cabinet in oblique projection.

2. Sketch given drawings in oblique projection.

3. Enjoy sketching drawings in oblique projection.

Key Inquiry Question:

What is the difference between cavalier and cabinet in oblique projection?

Learning Resources:

- Lesson notes

- Digital devices (tablets/laptops)

- Learner's books

- Pictures of cavalier and cabinet projections

- Drawing books

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson on basic projections and their importance in physics and engineering.

- Facilitate a brief discussion: Ask students to share any previous knowledge they may have regarding oblique projections.

- Introduce the key inquiry question: What is the difference between cavalier and cabinet projection?

Lesson Development (30 minutes):

Step 1: Introduction to Projections

- Use visual aids to present images of cavalier and cabinet projections.

- Explain the characteristics of each type, focusing on angles and proportions (cavalier typically maintains full height, while cabinet is half the height).

Step 2: Research Activity

- In pairs, students will use digital devices to search for additional information and examples of cavalier and cabinet projections.

- As students explore, circulate the room to provide guidance and answer any questions.

Step 3: Sketching Activity

- Each student will receive templates for simple objects (like a cube or a rectangular box) to sketch in both cavalier and cabinet projections.

- Encourage them to sketch freehand without using instruments, focusing on applying the principles discussed.

Step 4: Sharing and Peer Assessment

- Have students pair up with another pair to share their drawings and discuss the differences between their sketches.

- Encourage them to provide constructive feedback, referencing the characteristics of each projection type.

Conclusion (5 minutes):

- Summarize the key points: the characteristics of cavalier and cabinet projections, and the process of sketching them.

- Conduct a brief interactive poll: "Which projection do you prefer and why?" to encourage reflection and engagement.

- Prepare students for the next session by introducing how oblique projections relate to other types of projections they will learn in the future.

Extended Activities:

- Home Sketch Assignment: Students can take everyday objects from their home and attempt to sketch them in both cavalier and cabinet formats for further practice.

- Online Quiz: Create a Kahoot or Quizizz game focused on identifying the differences between cavalier and cabinet projection, including identifying sketches from their research.

Teacher Self-Evaluation:

WEEK 5: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub-Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Outline the steps for drawing shaped blocks in oblique projection using the cavalier projection.

2. Discuss the steps for drawing shaped blocks in oblique projection using cavalier projection.

3. Draw shaped blocks in oblique projection following the outlined steps.

4. Enjoy drawing the shaped blocks in oblique projection.

Key Inquiry Question(s):

- What are the key steps involved in creating an oblique projection of a shaped block?

- How does the cavalier projection differ from other types of projections?

Learning Resources:

- Whiteboard and markers

- Graph paper or drawing software

- Sample images of shapes with their oblique projections

- Handouts with a step-by-step guide for cavalier projection

- Projector (if applicable)

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson by asking students to recall what they learned about different types of projections and their importance in visualizing objects in 3D.

- Encourage a brief discussion on what they remember about shapes and projections.

Lesson Development (30 minutes):

Step 1: Outlining the Steps

- In small groups, learners discuss and outline the steps needed to draw a shaped block in cavalier projection.

- Each group should focus on identifying the main actions involved.

Step 2: Group Discussion

- Groups share their outlined steps with the class.

- Discuss differences and similarities between the steps outlined by each group.

- Clarify any misconceptions and finalize a comprehensive list of steps on the whiteboard.

Step 3: Demonstration

- The teacher demonstrates how to draw a simple shaped block (like a cube) using the cavalier projection technique on the board.

- Highlight key points in the drawing process, such as angle and proportion, and explain why these aspects are important.

Step 4: Drawing Practice

- Students turn to their individual papers and attempt to draw their own shaped block in oblique projection using the steps discussed.

- Encourage them to share their work with a partner for peer assessment and constructive feedback.

Conclusion (5 minutes):

- Summarize the key points discussed during the lesson, reinforcing the steps for drawing shaped blocks in oblique projection.

- Conduct a brief interactive activity, such as a quiz or a game, to reinforce the main learning points.

- Briefly preview the next lesson's focus on another type of projection or related concept to spark curiosity.

Extended Activities:

- Challenge students to find real-life objects that can be represented using cavalier projection and bring in pictures or sketches for the next lesson.

- Have students create a mini-project where they design and draw a complex object using oblique projection and present it to the class.

- Explore digital drawing tools or software that can be used for oblique projection to enhance their technical skills.

Teacher Self-Evaluation:

WEEK 6: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Draw shaped blocks in cavalier projection using drawing instruments.

2. Show interest in drawing shaped blocks in cavalier projection.

Key Inquiry Question:

- How do you draw a shaped block in cavalier projection using drawing instruments?

Learning Resources:

- Drawing books

- Learner's books

- Drawing instruments (e.g., rulers, compasses, protractors)

- Digital devices (tablets or computers)

Organisation of Learning:

Introduction (5 minutes):

1. Begin with a quick review of the previous lesson focused on basic geometry and projections.

2. Ask students to read a section on oblique projections from their drawing books.

3. Encourage a brief discussion on what they learned and share their thoughts on the importance of accurate drawing in design and engineering.

Lesson Development (30 minutes):

Step 1: Introduction to Cavalier Projection

- Explain what cavalier projection is and how it differs from other forms of drawing.

- Show visual examples of cavalier projections, outlining the characteristics such as the angle and scale of the depth.

Step 2: Preparing to Draw

- Distribute drawing instruments to each student (including pencils, rulers, and erasers).

- Demonstrate how to use the drawing instruments effectively, emphasizing the importance of precision in technical drawings.

Step 3: Drawing Activity

- In pairs or small groups, students will choose a simple shaped block (e.g., a cube, rectangular prism) and begin drawing it in cavalier projection on their paper.

- Circulate around the classroom to assist students as needed, and encourage them to apply the techniques discussed.

Step 4: Peer Review

- Once the drawings are complete, have students share their work with another pair/group.

- Encourage constructive feedback focusing on accuracy, use of instruments, and adherence to cavaliers' projection techniques.

Conclusion (5 minutes):

- Summarize the key points covered in the lesson, including what cavalier projection is and how to accurately represent shaped blocks.

- Conduct a brief interactive activity, such as a quiz or a quick drawing challenge based on the concepts learned.

- Preview the next session by asking students to think about how these drawing techniques apply to real-world engineering designs, setting the stage for future lessons.

Extended Activities:

- Homework Assignment: Ask students to find a real-world object, draw it in cavalier projection, and write a short paragraph explaining how the object’s dimensions relate to its function.

- Digital Exploration: Encourage students to use digital drafting tools (e.g., CAD software) if available, to practice creating cavalier projections of their drawings.

Teacher Self-Evaluation:

WEEK 6: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Outline the steps for drawing shaped blocks in cabinet projection.

2. Discuss the steps for drawing shaped blocks in cabinet projection.

3. Practice drawing shaped blocks in cabinet projection.

4. Develop a desire to draw shaped blocks in cabinet projection.

Key Inquiry Question:

- How do you draw shaped blocks in cabinet projection?

Learning Resources:

- Lesson notes
- Digital devices
- Video clips
- Samples of drawings in cabinet projection
- Learner's book
- Drawing books

Organisation of Learning:

Introduction (5 minutes):

- Begin with a quick review of the previous lesson on projection techniques.
- Ask students to read a brief section from their learner's book related to cabinet projection.
- Facilitate a brief discussion about any elements they found interesting or confusing, ensuring they engage with the key concepts.

Lesson Development (30 minutes):

Step 1: Observe Video Clip

- Show a video clip that demonstrates the process of drawing shaped blocks in cabinet projection.

- As they watch, encourage learners to take notes on any important steps or details they notice.

Step 2: Outline Steps

- After the video, guide each group to outline the steps they observed in the video.

- Provide a template for them to fill in, which can include sections for "Step #", "Action", and "Details".

Step 3: Discuss Outlined Steps

- Each group discusses their outlines with other groups, sharing their perspectives on the steps.

- Facilitate a class conversation to go through each outlined step to ensure understanding and clarity.

Step 4: Practice Drawing

- Provide drawing materials and allow learners to practice drawing a basic shaped block in cabinet projection using the steps outlined.

- Circulate around the room to provide assistance and feedback as needed.

Conclusion (5 minutes):

- Summarize the key points of the lesson, highlighting the steps for cabinet projection discussed.

- Conduct a quick interactive activity, such as a "pop quiz" where students raise their hands to answer questions about the drawing steps.

- Provide a preview of the next session, which could delve deeper into advanced projection methods or real-life applications of cabinet projection.

Extended Activities:

- Home Assignment: Create a drawing using cabinet projection of an object found in your home. Write a brief description of the drawing process and the steps followed.

- Group Project: Design a simple piece of furniture using cabinet projection techniques. Present the drawing and explain the rationale behind your design choices.

- Research Task: Investigate a famous architect or designer known for using 3D projections in their work and present your findings to the class.

Teacher Self-Evaluation:

WEEK 6: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, the learners should be able to:

1.Draw shaped blocks in cabinet projection using drawing instruments.

2.Enjoy drawing shaped blocks in cabinet projection using drawing instruments.

Key Inquiry Question(s):

- How do you draw shaped blocks in cabinet projection?

Learning Resources:

- Learner's books
- Drawing books
- Drawing instruments (rulers, protractors, compass, pencils)

Organisation of Learning:

Introduction (5 minutes):

- Begin with a brief review of the previous lesson. Ask students to share what they remember about oblique projections.
- Guide learners to read and discuss relevant content from their books, focusing on the concept of cabinet projection. Highlight key definitions and techniques.

Lesson Development (30 minutes):

Step 1: Understanding Cabinet Projection

- Explain the concept of cabinet projection. Discuss its importance in visualizing three-dimensional objects on a two-dimensional plane.
- Show examples of cabinet projection drawings and identify the key features.

Step 2: Introduction to Drawing Instruments

- Introduce the drawing instruments required for the task, such as rulers and protractors.
- Demonstrate how to properly use each instrument for drawing straight lines, creating angles, and ensuring precision.

Step 3: Drawing Practice

- In pairs or individually, instruct learners to select a shaped block (e.g., cube, cylinder).

- Guide them to use their instruments to create an accurate cabinet projection of their chosen shape on drawing paper. Assist them as needed.

Step 4: Peer Assessment and Feedback

- Allow students to share and display their drawings with the class or in small groups.

- Encourage peers to provide constructive feedback using guiding questions: What did you notice? How could it be improved?

Conclusion (5 minutes):

- Summarize the key points discussed and the learning objectives reached during the lesson.

- Conduct a brief interactive activity—ask students to identify one new thing they learned about cabinet projections and share it with the class.

- Preview the next session's topics, such as exploring other types of projections or practicing more complex shapes.

Extended Activities:

- Home Assignment: Ask students to find examples of objects in their homes and sketch how they would represent them in cabinet projection.

- Group Project: In small groups, students can create a poster exhibiting the different types of projections (isometric, oblique, etc.) and include examples and their applications in real life.

Teacher Self-Evaluation:

WEEK 6: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Oblique Projection

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. State the applications of oblique projection in drawing.

2. Discuss the uses of oblique drawings in our day-to-day life.

3. Walk around the locality to observe the use of oblique drawings.

4. Appreciate the applications of oblique projection in drawing.

Key Inquiry Question:

- How are oblique drawings used in technical fields?

Learning Resources:

- Learner's books
- Lesson notes
- Digital devices
- Surrounding environment

Organisation of Learning:

Introduction (5 minutes):

- Begin with a quick review of the previous lesson, asking students to recall key concepts related to different types of projections.
- Guide learners to read a selected passage from their books or lesson notes that discusses oblique projection, prompting a discussion that focuses on its key characteristics.

Lesson Development (30 minutes):

Step 1: Introduction to Oblique Projection

- Provide a clear definition of oblique projection, using visual aids to depict examples.
- Discuss the key features of oblique drawings, such as angles and dimensions, to illustrate how they differ from other projection types.

Step 2: Group Research Activity

- Divide students into small groups and assign them the task of using both digital devices and books to research specific applications of oblique projection.

- Each group should identify at least two professional fields (e.g., architecture, engineering) where oblique projection is significant.

Step 3: Class Discussion on Daily Life Uses

- Reconvene in the classroom and facilitate a discussion where each group shares their findings.

- Encourage students to think critically about how oblique drawings are used in everyday contexts, such as furniture design or urban planning.

Step 4: Local Observation Walk

- Take a brief walk around the locality, encouraging students to observe various structures or designs that may utilize oblique projection.

- Ask students to take notes or sketches of any examples they see and think about their practical implications.

Conclusion (5 minutes):

- Summarize the key points covered in the lesson, focusing on the practical applications and importance of oblique projection in both technical fields and everyday life.

- Host a brief interactive activity, such as a quiz or a reflection question where students share one new thing they learned about oblique projections.

- Prepare them for the next session by hinting at upcoming topics related to technical drawings and other projection methods.

Extended Activities:

- Research Project: Assign a short project where students can choose a specific field (like gaming, architecture, or engineering) and create a poster or slideshow about the use of oblique drawings in that field.

- Drawing Activity: Have students create their own oblique projection drawing of a simple object, such as a household item, showcasing their understanding of the concept.

- Real-World Comparison: Students can find examples of oblique versus isometric drawings and present their comparisons in small groups.

Teacher Self-Evaluation:

WEEK 7: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Visual Programming

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify application areas of visual programming software in solving problems.

2.Explain the application areas of visual programming software in solving problems.

3.Search the internet for information on application areas of visual programming.

4.Acknowledge the application areas of visual programming.

Key Inquiry Question(s):

- What are the application areas of visual programming?

Learning Resources:

- Lesson notes

- Computer learner's book

- Digital devices

Organisation of Learning:

Introduction (5 minutes):

- Begin with a review of the previous lesson. Ask students to recall the key points discussed.

- Introduce visual programming and its significance. Guide learners to read and discuss relevant content from the learning resources to set the stage for today's topic.

Lesson Development (30 minutes):

Step 1: Group Formation

- Divide the class into small groups (3-4 students each). Explain the objective of the group work and ensure each group has access to either printed resources or digital devices.

Step 2: Research

- In their groups, students will use print or digital resources to search for information on visual programming. Encourage them to focus on various applications such as game design, simulations, data visualization, and educational tools.

Step 3: Identification

- Each group will compile a list of application areas they discovered during their research. Instruct them to write down a short description of how each application area uses visual programming.

Step 4: Discussion

- Groups will present their findings to the class. Encourage discussion after each presentation, asking other groups to contribute any additional information or applications they may have discovered.

Conclusion (5 minutes):

- Summarize the key points discussed throughout the lesson: what visual programming is and its various applications.

- Conduct a brief interactive activity, such as a quiz or brainstorming session, to reinforce the main topics. Invite learners to ask questions for clarification.

- Preview the next lesson, hinting at what will be covered and encouraging students to think about how they use technology in daily life.

Extended Activities:

- Project Opportunity: Assign students to create a simple visual programming project that highlights one application area they researched. This could be a small game or a visual simulation.

- Research Paper: Have students write a one-page summary on the impact of visual programming in the field of their interest (such as science, technology, or the arts), encouraging them to delve deeper into a specific application.

- Guest Speaker: Invite a local software developer or educator to speak about their experiences using visual programming in their profession.

Teacher Self-Evaluation:

WEEK 7: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Visual Programming

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Outline the steps for creating an application using visual programming software for solving day-to-day problems.

2. Use digital devices to search for clips or information on creating an application using visual programming software.

3. Express interest in developing applications using visual programming software.

Key Inquiry Question(s):

- How are applications developed using visual programming software?

Learning Resources:

- Lesson notes
- Video clips
- Digital devices (tablets/computers)
- Computer learner's book

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson on programming basics.
- Discuss the importance of visual programming and applications in everyday life.
- Guide learners to read and discuss relevant sections from the lesson notes, highlighting key concepts related to visual programming.

Lesson Development (30 minutes):

Step 1: Research

- Activity: In groups, learners will use digital devices to search for and watch video clips on developing applications using visual programming software. They will focus on examples like games, stories, and animations.
- Outcome: Understand different application types that can be created using visual programming.

Step 2: Outline Steps

- Activity: Based on the video clips watched, each group will outline the major steps involved in creating an application using visual programming software.

- Discussion Points: What tools are used? What are the important phases (ideation, design, testing)?

Step 3: Detailed Discussion

- Activity: Groups will discuss their outlined steps with the class, sharing insights on what each step involves and its significance.

- Outcome: Collaborative learning and feedback on the outlined steps.

Step 4: Practical Application

- Activity: Each group will brainstorm an application idea and draft a simple plan of what they would like to create if they had access to visual programming software.

- Outcome: Encourage creativity and application of learned concepts.

Conclusion (5 minutes):

- Summarize key points discussed and review the steps needed to create an application.

- Conduct a quick interactive poll: "What application would you like to develop and why?"

- Preview the next session: Introduction to specific visual programming tools and languages.

Extended Activities:

- Activity Suggestion:

- Encourage students to pick one of the discussed applications (like a simple game or story) and create a flowchart of how they would program it step-by-step.

- Optional Homework: Research and present on a famous application developed using visual programming (e.g., Scratch, App Inventor).

Teacher Self-Evaluation:

WEEK 7: LESSON 3 - 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Visual Programming

Specific Learning Outcomes:

- By the end of the lesson, the learner should be able to:

1.Create an application using visual programming software for solving problems in daily life.

2.Learners will enjoy creating applications using the visual programming software.

Key Inquiry Question:

- How are applications developed using visual programming software?

Learning Resources:

- Digital devices (tablets/computers)

- Video clips demonstrating visual programming

- Projector

- Computer learner's book

Organisation of Learning:

Introduction (5 minutes):

- Review the Previous Lesson: Start by quick questioning to recap what students learned in the last class regarding the basics of programming concepts.

- Discuss Key Concepts: Guide students as they read from the learning resources, focusing on the features and importance of visual programming software. Encourage participation and discussion.

Lesson Development (30 minutes):

Step 1: Introduction to Visual Programming Software

- Present a short video clip showcasing different applications created using visual programming software. Highlight how easy and fun it is to use these tools.

- Discuss the benefits of using visual programming, such as its user-friendly interface.

Step 2: Brainstorming Session

- Organize students into small groups.

- Have each group brainstorm a problem from daily life that they can address with an application. Encourage them to think creatively and share ideas within their group.

Step 3: Application Development

- Using digital devices, allow each group to begin creating their interactive stories, games, or animations based on their chosen problem.

- Circulate and provide guidance, ensuring students understand how to utilize the features of the software.

Step 4: Presentation and Feedback

- Invite each group to briefly share their creations with the class. Encourage positive feedback from peers on how each application addresses their specific problem.

- Highlight creativity and various solutions depicted by different groups.

Conclusion (5 minutes):

- Summarize Key Points: Recap the learning objectives and successes of the lesson, such as the creation of applications and collaborative learning.

- Interactive Activity: Engage students in an interactive quiz or game related to concepts learned during the lesson to reinforce their understanding.

- Preview Next Session: Briefly discuss what students will learn next, such as further exploration of advanced features in visual programming software or related physics concepts.

Extended Activities:

- Homework Assignment: Students can identify a problem at home or in their community and draft a plan for an application they would create to solve it, detailing the features they would include.

- After-School Club: Encourage interested students to join a coding club where they can further explore ideas and collaborate on more complex projects using visual programming software.

Teacher Self-Evaluation:

WEEK 9: LESSON 1 - 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Communication in Pre-Technical Studies

Sub Strand: Visual Programming

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Develop interactive stories, games, and animations using visual programming software.

2. Embrace the use of visual programming in their daily life.

Key Inquiry Questions:

- What types of visual programming applications can you use to create games, stories, and animations?

Learning Resources:

- Lesson notes

- Digital devices: Laptops, Tablets

- Video clips explaining visual programming concepts

Organisation of Learning:

Introduction (5 minutes):

- Begin with a brief review of the previous lesson on programming concepts.

- Engage students in a discussion about what they know about visual programming. Ask guiding questions to draw connections to their everyday experiences (e.g., games they play).

Lesson Development (30 minutes):

Step 1: Introduction to Visual Programming

- Present video clips that demonstrate various visual programming applications (like Scratch).

- Highlight examples of stories, games, and animations made with visual programming.

Step 2: Explore the Software

- Students will access visual programming software (Scratch or Sprite) on their devices.

- Each student or pair will explore the interface, identifying key features such as sprites, scripts, stages, and backdrops.

Step 3: Create Interactive Projects

- Guide students to brainstorm ideas for a simple interactive story or game they would like to build.

- Encourage them to sketch a plan on paper before starting to code in the software.

Step 4: Present and Share Creations

- Allow each pair or individual to present their projects to the class.

- Invite feedback from peers and discuss how they used visual programming to convey their ideas.

Conclusion (5 minutes):

- Summarize key points about visual programming learned during the lesson (e.g., the significance of programming in creating games).

- Conduct a brief interactive quiz using a tool like Kahoot! to reinforce concepts discussed.

- Preview upcoming sessions that may cover more advanced programming concepts or environmental applications of programming.

Extended Activities:

- Game Design Challenge: Encourage students to create a more complex game with at least three levels over the weekend and present it in the next class.

- Story Development Workshop: Ask students to write a short story and then animate it using the visual programming software. They can incorporate dialogue and sound effects.

- Community Improvement Project: Students could brainstorm how visual programming could be applied to solve a real-world problem in their community and create a presentation for it.

Teacher Self – Evaluation:

WEEK 9: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Wood

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the types of wood used in production.

2. Use digital or print resources to search for information on the types of wood used in production.

3. Acknowledge the different types of wood used in production.

Key Inquiry Question(s):

- What types of wood are used in production?

Learning Resources:

- Lesson notes

- Learner's textbook

- Digital devices (tablets/computers)

- Pictures of different wood types

Organisation of Learning:

Introduction (5 minutes):

- Begin with a quick review of the previous lesson on materials used in production.

- Engage learners in a discussion by asking questions about wood and its importance in production.

- Guide learners to read a segment from the lesson notes and textbook related to wood types.

Lesson Development (30 minutes):

Step 1: Group Formation

- Divide learners into small groups of 4-5.

- Assign roles within the group, such as note-taker, researcher, and presenter.

Step 2: Guided Research

- Instruct groups to use their digital devices or print resources to search for information regarding various types of wood (e.g., hardwoods like oak and softwoods like pine).

- Encourage them to note down characteristics, uses, and examples of each type.

Step 3: Group Discussion

- Have groups come together to discuss their findings. They should share the information about the types of wood they researched and how it’s used in production.

- Ask each group to prepare a brief presentation on one type of wood, highlighting its features and applications.

Step 4: Presentations

- Each group presents their findings to the class.

- Encourage questions and discussions after each presentation to reinforce understanding.

Conclusion (5 minutes):

- Summarize the key points learned about different types of wood and their uses in production.

- Conduct a quick interactive activity, such as a matching game where learners match wood types with their characteristics or uses.

- Preview the topic for the next session, possibly introducing materials made from these woods or the process of wood treatment.

Extended Activities:

- Wood Gallery Creation: Ask learners to create a poster or digital slideshow on a specific type of wood that includes images, properties, and applications.

- Field Trip Proposal: Consider organizing a visit to a local carpentry workshop or lumber yard to see wood types used in production firsthand.

- Research Project: Assign a longer-term project where learners can choose a type of wood and conduct a detailed research report on its history, properties, and common uses in various industries.

Teacher Self-Evaluation:

WEEK 9: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Wood

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the physical characteristics of softwood and hardwood used in production.

2. Use digital devices to search for information on the physical characteristics of softwood and hardwood.

3. Acknowledge the physical characteristics of softwood and hardwood that make them suitable for production.

Key Inquiry Question(s):

- What are the physical characteristics of hard and soft wood?

Learning Resources:

- Lesson notes

- Learner's Textbook

- Digital devices (computers, tablets, etc.)

- Posters and flashcards

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson about materials used in production.

- Prompt learners to recall what they learned about wood.

- Guide them to read and discuss relevant content from the lesson notes and textbooks, focusing on the physical characteristics of wood.

Lesson Development (30 minutes):

Step 1: Group Formation

- Divide the class into small groups of 4-5 students to encourage collaboration.

Step 2: Research Phase

- In their groups, learners will use digital devices to search for information relating to the physical characteristics of softwood and hardwood.

- Provide guiding questions such as: What does softwood look like? What are the properties of hardwood?

Step 3: Identification and Discussion

- Groups share their findings with each other, focusing on identifying and discussing the physical characteristics of each type of wood.

- Encourage them to consider features such as density, texture, color, and usage in production.

Step 4: Creation of Visual Aids

- Using the information gathered, each group will create a poster or flashcard that displays the physical characteristics of hard and soft wood.

- Allow them to be creative in their presentations, including illustrations and key characteristics.

Conclusion (5 minutes):

- Summarize the key points discussed during the lesson: the differences between softwood and hardwood, and their respective physical characteristics.

- Conduct a brief interactive activity, such as a “wood characteristics quiz” where students can answer questions based on their posters.

- Lastly, preview the next lesson's topic, which could be about the applications of wood in various industries.

Extended Activities:

- Field Trip: Organize a visit to a local wood processing plant or a natural reserve where different types of trees grow, letting students observe real-life applications of softwood and hardwood.

- Creative Project: Have students create a model or diorama showing the different habitats where softwood and hardwood trees might be found, including descriptions of the trees' characteristics and the types of products they produce.

- Research Report: Students can write a short report comparing the environmental impact of using softwood vs. hardwood in construction and manufacturing.

Teacher Self-Evaluation:

WEEK 10: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Wood

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the physical properties used in classifying woods used in production.

2. Classify wood according to the physical properties.

3. Enjoy classifying the different types of wood used in material production.

Key Inquiry Question(s):

- What criteria will you use to classify the different types of wood used in material production?

Learning Resources:

- Lesson notes

- Learner's Textbook

- Digital devices (tablets or computers if available)

- Classification tables

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson about materials used in production.

- Encourage students to share what they remember and relate it to today's focus on wood.

- Guide learners to read and discuss relevant content from the learning resources, emphasizing key physical properties of wood (density, grain, color, texture).

Lesson Development (30 minutes):

Step 1: Identifying Physical Properties

- In groups, students will brainstorm and list physical properties used in classifying wood (e.g., hardness, color, weight).

- Each group will present their list to the class, and the teacher will compile a master list on the board.

Step 2: Classifying Woods

- Guide students to understand the difference between softwood and hardwood, focusing on their physical properties.

- Distribute digital devices for learners to look up examples of both types of wood and review their characteristics.

Step 3: Sorting Activity

- Provide students with a checklist of physical properties.

- Each group will receive samples or pictures of different woods and will use the checklist to sort them into softwood and hardwood categories.

Step 4: Presenting Classification

- Groups will create a classification table of wood according to the physical properties identified.

- Each group will present their classification table to the rest of the class, explaining their reasoning for their choices.

Conclusion (5 minutes):

- Summarize key points about the classification of wood based on physical properties and discuss how these characteristics affect material production.

- Conduct a brief interactive activity where students guess the type of wood based on its properties, to reinforce the main topics.

- Prepare learners for the next session by previewing content about the uses of different types of wood in production, encouraging them to think of different examples in their everyday life.

Extended Activities:

- Wood Investigation Project: Have students select a type of wood not covered in class. They will research its properties, uses, and sustainability and present their findings in a short report or poster.

- Field Trip: Arrange a visit to a local lumber yard or woodworking shop to see different types of wood and how they are processed and used.

- Creative Project: Encourage students to design a simple product (like a birdhouse or shelf) and describe the wood they would choose for it based on physical properties, including reasons for their choices.

Teacher Self-Evaluation:

WEEK 10: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Wood

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Define the term "seasoning" as used in wood preparation.

2. Describe various seasoning methods in wood preparation.

3. Search the internet for clips on the seasoning methods of wood.

4. Acknowledge the need for wood seasoning in wood preparation.

Key Inquiry Questions:

- What are the benefits of wood seasoning?

- What is seasoning as a method of wood preparation?

Learning Resources:

- Lesson notes

- Digital devices (tablets/computers)

- Learner's Textbook

- Video clips (tutorials on wood seasoning)

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson, focusing on materials used in production.

- Highlight the importance of wood in various applications.

- Guide learners to read and discuss relevant content from the learning resources, emphasizing key concepts such as "seasoning" and its significance.

Lesson Development (30 minutes):

Step 1: Define Seasoning

- In groups, learners will brainstorm the concept of "seasoning" in wood preparation.

- Each group will share their definitions and ideas with the class.

- Teacher will summarize and refine the definitions based on student feedback.

Step 2: Investigate Seasoning Methods

- Using digital or print resources, groups will research different methods of wood seasoning.

- Encourage learners to find examples of air drying, kiln drying, and other methods.

- Groups will prepare to present their findings to the class.

Step 3: Discover Benefits of Seasoning

- Learners will discuss the importance of seasoning in wood preparation.

- Questions to consider: Why is seasoning necessary? What problems can occur if wood is not seasoned?

Step 4: Multimedia Exploration

- Use digital devices to search for and watch video clips on wood seasoning methods.

- Groups will share key takeaways from the videos with the class.

Conclusion (5 minutes):

- Summarize the key points discussed in the lesson: definition of seasoning, methods, and benefits.

- Conduct a brief interactive quiz or activity to reinforce the main topics addressed during the lesson.

- Prepare learners for the next session by previewing upcoming topics, such as the physical properties of seasoned wood.

Extended Activities:

-Wood Seasoning Experiment: Learners can conduct a simple experiment at home by seasoning small pieces of wood in different environments (air vs. a controlled setting). They can record changes and present their findings.

-Research Project: Encourage students to choose a specialty wood (e.g., oak, pine) and research its seasoning process, including traditional and modern methods. Students can create a presentation to share with the class.

- Field Trip: If possible, arrange a visit to a local sawmill or woodworking shop where students can see the wood seasoning process in action and ask questions.

Teacher Self-Evaluation:

WEEK 10: LESSON 3

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Wood

Specific Learning Outcomes:

- By the end of the lesson, learners will be able to:

1. Relate various types of wood to their commonly known uses in the community.
2. Discuss the uses of different types of wood within their locality.
3. Develop charts depicting the types of wood and their specific uses.
4. Value the importance of wood in daily life.

Key Inquiry Question:

- Why is wood important in day-to-day life?

Learning Resources:

- Lesson notes
- Digital devices (tablets/computers)
- Charts
- Metre rule
- Marker pens
- Learner's textbook

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson by asking students to recall key concepts related to materials used in construction and products.

- Guide learners to read and discuss relevant content from the learning resources, emphasizing the understanding of different types of wood and their properties.

Lesson Development (30 minutes):

Step 1: Brainstorming Session

- In groups, learners will brainstorm on the uses of wood in various trades such as furniture making, construction, and artwork.

- Each group will list their ideas on a piece of paper or digital device, focusing on local examples.

Step 2: Matching Wood Types to Uses

- Provide learners with a list of types of wood (e.g., oak, pine, cedar) and ask groups to match each type to its specific use.

- Encourage them to discuss why certain woods are better suited for particular purposes based on characteristics like strength, durability, and cost.

Step 3: Chart Preparation

- Learners will work together to prepare charts that illustrate the types of wood they discussed along with their uses.

- Each chart should include visuals (drawings or printed images) and brief explanations.

Step 4: Local Exploration Follow-up

- Discuss potential locations in the community where learners can explore wood uses (e.g., local carpenters, furniture stores, parks).

- Plan an upcoming field trip or a project where they observe the uses of wood in real life.

Conclusion (5 minutes):

- Summarize key points from the lesson: types of wood, their uses, and their importance in daily life.

- Conduct an interactive activity, such as a quick quiz or think-pair-share, to reinforce main topics discussed.

- Prepare learners for the next session by introducing the upcoming topic on sustainable wood sourcing or conservation.

Extended Activities:

- Wood Characteristics Research: Assign learners to research a specific type of wood and present on its properties, sustainability, and uses.

- Creative Project: Have students create a model or a small project using different types of wood, illustrating their learning on how different woods are used in everyday objects.

- Field Trip Report: If a field trip is conducted, learners can write a report or create a presentation on what they learned about the local uses of wood.

Teacher Self-Evaluation:

WEEK 10: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Wood

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the methods used in wood preparation.
2. Describe the conversion method as a method of wood preparation.
3. Search the internet for information and clips on the conversion method of wood.
4. Appreciate the conversion method in wood preparation.

Key Inquiry Question(s):

- What is the importance of the conversion method in wood preparation?

Learning Resources:

- Lesson notes
- Digital devices (tablets/laptops)
- Learner's Textbook
- Video clips

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson by asking students what they remember about wood preparation.

- Guide learners to read and discuss relevant content from their textbook or lesson notes, emphasizing the significance of understanding wood preparation methods.

Lesson Development (30 minutes):

Step 1: Brainstorming

- In small groups, learners discuss and brainstorm all the different methods of wood preparation they know or have learned about.

- Each group will list at least three methods and prepare to present them.

Step 2: Research on Conversion Method

- Using digital devices, groups will search for information and video clips that explain the conversion method of wood preparation.

- Encourage learners to focus on key factors such as the purpose, processes, advantages, and any innovative technologies used in this method.

Step 3: Group Discussion

- Each group shares their findings on the conversion method with the class.

- Foster a class discussion by asking guiding questions like: "Why is the conversion method essential in wood preparation?"

Step 4: Reflection and Appreciation

- Let learners discuss in pairs the importance of the conversion method and how it ties into sustainable practices in wood production.

- Wrap up this section by asking for shared thoughts on how technology may enhance wood preparation.

Conclusion (5 minutes):

- Summarize key points from the lesson, reiterating the importance of wood preparation methods and the conversion method specifically.

- Conduct a quick interactive activity, like a thumbs-up/thumbs-down polling, to gauge understanding of the main topics discussed.

- Provide a preview of the next session, hinting at discussions about sustainability in wood usage.

Extended Activities:

- Research Project: Ask students to choose a specific wood preparation method or technology and create a short presentation or poster to share with the class.

- Field Trip/Virtual Tour: If possible, plan a visit to a local sawmill or a virtual tour of a wood processing facility to see the conversion method in action.

- Creative Writing: Invite students to write a short story or reflection on the journey of a piece of wood from tree to finished product, incorporating what they've learned about preparation methods.

Teacher Self-Evaluation:

WEEK 11: LESSON 1

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Handling Waste Materials

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the waste materials found in the environment.
2. Collect waste products in the school compound.
3. Sort the collected waste materials according to their types.
4. Enjoy the process of collecting and sorting different types of waste products.

Key Inquiry Question(s):

- Which waste materials are found in the environment?

Learning Resources:

- Lesson notes
- Digital devices
- School Environment
- Cartons
- Sacks

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson about materials used in production and their lifecycle.

- Guide learners to read from the lesson notes, focusing on identifying waste materials. Facilitate a brief discussion to ensure understanding of key concepts related to waste handling.

Lesson Development (30 minutes):

Step 1: Identification

- In groups, learners will walk around the school compound with their digital devices or notebooks to identify various waste materials such as plastic bottles, paper, metal cans, and organic waste. Each group should make a list of what they find.

Step 2: Collection

- Using sacks and cartons provided, students will collect the identified waste materials. Remind them to be careful and follow safety guidelines while handling items.

Step 3: Sorting

- Back in the classroom or designated sorting area, groups will sort the collected waste materials into categories: plastic, glass, metal, wood, electronic, and construction waste. Provide clear labels for each category.

Step 4: Storage

- Once sorted, learners will store the waste materials in the appropriate cartons or sacks, ensuring that each type of waste is kept separate. Discuss briefly why sorting is important in waste management.

Conclusion (5 minutes):

- Summarize the key points: types of waste, importance of sorting, and the process of collection.

- Conduct a brief interactive activity, like a quiz or a trash-related trivia game, to reinforce what they learned.

- Preview the next session by posing questions such as, "What happens to waste after it is sorted?” or “Why is it important to recycle?”

Extended Activities:

- Research Project: Learners can choose a type of waste and research its environmental impact, possible recycling methods, and ways to reduce its consumption in daily life. Present findings in a creative format like a poster or infographic.

- Community Clean-Up: Organize a school or community event where students can clean a local park or area, applying their learning in a real-world setting while fostering community responsibility.

- Guest Speaker: Invite someone from a local recycling company or environmental organization to discuss the importance of waste management and recycling in depth.

Teacher Self-Evaluation:

WEEK 11: LESSON 2

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Handling Waste Materials

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the ways of handling waste materials safely in the environment.

2. Describe ways of handling waste materials safely in the environment.

3. Search the internet for information on safe ways of handling waste materials.

4. Acknowledge the safe ways of handling waste materials in the environment.

Key Inquiry Question:

- How are waste materials handled in the locality?

Learning Resources:

- Lesson notes

- Digital devices (tablets, laptops)

- Learner's Textbook

Organisation of Learning:

Introduction (5 minutes):

- Review the previous lesson on environmental impacts of waste.

- Guide learners to read and discuss relevant content on waste management from the textbook and lesson notes, emphasizing the importance of safe waste handling.

Lesson Development (30 minutes):

Step 1: Brainstorming

- In small groups, learners will discuss and list different waste materials commonly found in their locality.

- Each group will identify how these materials are currently being disposed of or recycled.

Step 2: Research

- Groups will use digital devices to search for information online about safe methods of handling and disposing of these waste materials.

- Encourage learners to look for local initiatives or organizations that focus on waste management.

Step 3: Discussion

- Each group will present their findings and discuss the safe ways of handling waste materials they discovered in their research.

- Facilitate a class discussion on the effectiveness and importance of these methods.

Step 4: Reflection

- Ask groups to reflect on how their community handles waste materials and suggest improvements based on what they learned.

- This could be written down or shared with the class.

Conclusion (5 minutes):

- Summarize the key points covered in the lesson, reinforcing the importance of safe waste handling practices.

- Conduct a brief interactive activity, such as a quiz or group reflection, to reinforce the main topics.

- Prepare students for the next session by previewing upcoming topics related to waste recycling and environmental sustainability.

Extended Activities:

- Community Waste Audit: Students can conduct a small waste audit in their homes or schools, documenting the types of waste produced and proposing recycling or disposal solutions.

- Create a Waste Management Plan: Learners can work individually or in groups to create a waste management plan for their classroom or school, including methods for reducing waste, reusing materials, and recycling.

- Awareness Campaign: Students can develop a poster or presentation that highlights the importance of handling waste safely, which can be displayed around the school or shared with the community.

Teacher Self-Evaluation:

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL

	GRADE 9	**PRE - TECHNICAL STUDIES**			

WEEK 11: LESSON 3

Strand: Materials for Production

Sub Strand: Handling Waste Materials

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. Identify the methods of disposing of waste materials in the environment.

2. Dispose of waste materials using appropriate methods.

3. Acknowledge the need for disposing of waste materials appropriately in the environment.

Key Inquiry Question(s):

- How can we dispose of waste materials in the environment?

Learning Resources:

- Lesson notes

- Digital devices

- School environment

- School dustbin

Organisation of Learning:

Introduction (5 minutes):

- Begin the lesson by reviewing the previous topics covered on materials and their environmental impact.

- Prompt a discussion by asking students how they currently dispose of their waste and what types of waste they notice in their surroundings.

- Guide learners to read and discuss relevant content from the provided lesson notes with a focus on understanding disposal methods.

Lesson Development (30 minutes):

Step 1: Understanding Waste Types

- In groups, have learners discuss and categorize different types of waste materials (e.g., organic, plastic, paper, metal).

- Each group presents the categories they identified, which fosters listening and speaking skills.

Step 2: Research Disposal Methods

- Assign each group a specific disposal method: recycling, composting, reusing, and incinerating (burning).

- Allow learners to use digital devices to research their assigned method, focusing on how it works and its benefits.

Step 3: Plan for School Environment

- Groups will brainstorm and create a brief action plan to implement their findings within the school environment, focusing on encouraging their peers to practice proper waste disposal.

Step 4: Demonstration & Practice

- Conduct a safe disposal demonstration in the school environment using the school dustbin.

- Allow students to practice proper disposal techniques with real waste materials under supervision.

Conclusion (5 minutes):

- Recap the key points discussed in each step of the lesson, emphasizing the importance of proper waste management methods.

- Conduct an interactive quiz or a quick reflection activity where students share one new thing they learned about waste disposal.

- Preview the next lesson's topic on environmental impact and pollution, encouraging students to think about how waste affects their community.

Extended Activities:

- Waste Audit Project: Have students conduct an audit of the types of waste generated at home or in school. They can present their findings and suggest improvement strategies.

- Community Campaign: Encourage learners to design a campaign promoting proper waste disposal techniques within their community, including posters and social media posts.

- Field Trip: Plan a visit to a recycling plant or a composting facility to give students a firsthand look at waste management in action.

Teacher Self-Evaluation:

WEEK 11: LESSON 4

SCHOOL	LEVEL	LEARNING AREA	DATE	TIME	ROLL
	GRADE 9	PRE - TECHNICAL STUDIES			

Strand: Materials for Production

Sub Strand: Handling Waste Materials

Specific Learning Outcomes:

- By the end of the lesson, learners should be able to:

1. State the importance of proper handling of waste materials in the environment.

2. Search the internet or print resources for information on the importance of proper handling of waste materials in the environment.

3. Appreciate the need for proper handling of waste materials in the environment.

Key Inquiry Question:

- Why is it important to handle waste materials properly in the environment?

Learning Resources:

- Lesson notes

- Learner's Textbook

- Digital devices (tablets, laptops)

- Posters

Organisation of Learning:

Introduction (5 minutes):

- Begin the lesson by reviewing key concepts from the previous lesson on materials and their properties.

- Ask students to share ideas on what waste materials are and where they can commonly be found.

- Transition to the topic of waste materials and their handling by highlighting its relevance to the environment.

Lesson Development (30 minutes):

Step 1: Brainstorming

- Divide the class into small groups.

- Prompt each group to brainstorm and list as many reasons as possible for why proper handling of waste materials is important.

- Encourage them to think about environmental, health, and economic impacts.

Step 2: Research

- Equip each group with digital devices or print resources.

- Instruct them to research specific examples and detailed reasons for the importance of correct waste management.

- Encourage students to look for statistics and case studies that can add depth to their findings.

Step 3: Discussion

- Bring the class back together and have each group share key points from their research.

- Facilitate a class discussion that highlights common themes and ideas discovered during their research.

Step 4: Poster Preparation

- Instruct each group to create a simple poster summarizing their findings.

- Encourage creativity, and ask them to include graphics or illustrations that can help convey the importance of proper waste handling.

Conclusion (5 minutes):

- Summarize the key points discussed during the lesson and emphasize the importance of proper waste management for the environment.

- Conduct a brief interactive activity such as a "thumbs up/thumbs down" vote to gauge student understanding of the concepts discussed, or have them reflect on a question: "What will you do differently about waste in your community?"

- Provide a preview of the next session, which will focus on waste management techniques and practices.

Extended Activities:

- Research Project: Assign learners to choose a specific waste material (like plastic or paper) and research its environmental impact, including how it can be recycled or reused effectively. Present their findings in a short report or presentation.

- Community Clean-Up: Encourage students to participate in or organize a local clean-up event at a nearby park or school. Have them document their experience and reflect on the importance of waste management in practice.

Teacher Self-Evaluation: